Team 5
Presentation 2

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Parquet Table (Cases Summary

Case-Level Summary

First 20 cases · label: red = 1, blue = 0

case_id	patient_id	label	grade	fold	n_patches	n_slices	stains	n_missing_emb	note
1	1	1	4	F2	452	4	H&E, Melan, SOX10	0	—
2	2	0	1	F3	673	4	H&E, Melan, SOX10	0	—
3	3	1	4	F4	862	4	H&E, Melan, SOX10	1	—
4	4	0	1	F5	941	3	H&E, Melan, SOX10	**37**	—
5	5	1	3	F1	508	2	H&E, Melan, SOX10	5	—
7	7	1	3	F3	502	3	H&E, Melan, SOX10	0	—
9	9	1	3	F5	226	3	H&E, Melan, SOX10	0	—
11	11	1	3	F2	575	4	H&E, Melan, SOX10	2	—
12	12	1	3	F1	424	3	H&E, Melan, SOX10	2	—
14	14	1	4	F4	912	7	H&E, Melan, SOX10	0	—
15	15	1	3	F3	2,132	14	H&E, Melan, SOX10	0	—
16	16	1	4	F5	177	1	H&E	0	—
17	17	1	4	F2	502	3	H&E, SOX10	**20**	—
18	18	1	3	F1	960	5	H&E, Melan, SOX10	**34**	—
19	19	1	3	F4	2,445	8	H&E, Melan, SOX10	**119**	—
21	21	0	1	F2	710	5	H&E, Melan	0	—
22	22	0	1	F4	745	6	H&E, Melan, SOX10	4	—
23	23	0	1	F1	988	5	H&E, Melan, SOX10	3	—
24	24	0	1	F3	945	6	H&E, Melan, SOX10	0	—
25	25	0	1	F5	467	3	H&E, Melan, SOX10	0	—

Red text marks missing embeddings (bold ≥ 20) and cases missing one or more stains.

- One row per case (**101 rows**)
- Shows the case of every state
- Easily lets us know the gaps for each case
- Splits are easy to check.
- Because of how small it is, very easy to share.

Javi

Parquet Table Patches

patch_id	case	patient	slice_raw	slice_id	stain	patch	png_name	png_path	emb_path	exclude_reason	grade_raw	label	fold	emb_version	dataset	built_at
case_1_match_1_h&e_patch0	1	1	match_1	match_1	h&e	0	case_1_match_1_h&e_patch0.png	/projects/e32998/pa...	/projects/e32998/patches_varsize_poole...		4	1	F2	kimianet_pool2x2	v1	2026-11-05T...
case_1_match_1_h&e_patch1	1	1	match_1	match_1	h&e	1	case_1_match_1_h&e_patch1.png	/projects/e32998/pa...	/projects/e32998/patches_varsize_poole...		4	1	F2	kimianet_pool2x2	v1	2026-11-05T...
case_1_match_1_h&e_patch2	1	1	match_1	match_1	h&e	2	case_1_match_1_h&e_patch2.png	/projects/e32998/pa...	/projects/e32998/patches_varsize_poole...		4	1	F2	kimianet_pool2x2	v1	2026-11-05T...
case_1_match_1_h&e_patch3	1	1	match_1	match_1	h&e	3	case_1_match_1_h&e_patch3.png	/projects/e32998/pa...	/projects/e32998/patches_varsize_poole...		4	1	F2	kimianet_pool2x2	v1	2026-11-05T...
case_1_match_1_h&e_patch4	1	1	match_1	match_1	h&e	4	case_1_match_1_h&e_patch4.png	/projects/e32998/pa...	/projects/e32998/patches_varsize_poole...		4	1	F2	kimianet_pool2x2	v1	2026-11-05T...
case_1_match_1_h&e_patch5	1	1	match_1	match_1	h&e	5	case_1_match_1_h&e_patch5.png	/projects/e32998/pa...	/projects/e32998/patches_varsize_poole...		4	1	F2	kimianet_pool2x2	v1	2026-11-05T...
case_1_match_1_h&e_patch6	1	1	match_1	match_1	h&e	6	case_1_match_1_h&e_patch6.png	/projects/e32998/pa...	/projects/e32998/patches_varsize_poole...		4	1	F2	kimianet_pool2x2	v1	2026-11-05T...
case_1_match_1_h&e_patch7	1	1	match_1	match_1	h&e	7	case_1_match_1_h&e_patch7.png	/projects/e32998/pa...	/projects/e32998/patches_varsize_poole...		4	1	F2	kimianet_pool2x2	v1	2026-11-05T...
case_1_match_1_h&e_patch8	1	1	match_1	match_1	h&e	8	case_1_match_1_h&e_patch8.png	/projects/e32998/pa...	/projects/e32998/patches_varsize_poole...		4	1	F2	kimianet_pool2x2	v1	2026-11-05T...
case_1_match_1_h&e_patch9	1	1	match_1	match_1	h&e	9	case_1_match_1_h&e_patch9.png	/projects/e32998/pa...	/projects/e32998/patches_varsize_poole...		4	1	F2	kimianet_pool2x2	v1	2026-11-05T...
case_1_match_1_h&e_patch10	1	1	match_1	match_1	h&e	10	case_1_match_1_h&e_patch10.png	/projects/e32998/pa...	/projects/e32998/patches_varsize_poole...		4	1	F2	kimianet_pool2x2	v1	2026-11-05T...
case_1_match_1_h&e_patch11	1	1	match_1	match_1	h&e	11	case_1_match_1_h&e_patch11.png	/projects/e32998/pa...	/projects/e32998/patches_varsize_poole...		4	1	F2	kimianet_pool2x2	v1	2026-11-05T...
case_1_match_1_h&e_patch12	1	1	match_1	match_1	h&e	12	case_1_match_1_h&e_patch12.png	/projects/e32998/pa...	/projects/e32998/patches_varsize_poole...		4	1	F2	kimianet_pool2x2	v1	2026-11-05T...
case_1_match_1_h&e_patch13	1	1	match_1	match_1	h&e	13	case_1_match_1_h&e_patch13.png	/projects/e32998/pa...	/projects/e32998/patches_varsize_poole...		4	1	F2	kimianet_pool2x2	v1	2026-11-05T...
case_1_match_1_h&e_patch14	1	1	match_1	match_1	h&e	14	case_1_match_1_h&e_patch14.png	/projects/e32998/pa...	/projects/e32998/patches_varsize_poole...		4	1	F2	kimianet_pool2x2	v1	2026-11-05T...
case_1_match_1_h&e_patch15	1	1	match_1	match_1	h&e	15	case_1_match_1_h&e_patch15.png	/projects/e32998/pa...	/projects/e32998/patches_varsize_poole...		4	1	F2	kimianet_pool2x2	v1	2026-11-05T...
case_1_match_1_h&e_patch16	1	1	match_1	match_1	h&e	16	case_1_match_1_h&e_patch16.png	/projects/e32998/pa...	/projects/e32998/patches_varsize_poole...		4	1	F2	kimianet_pool2x2	v1	2026-11-05T...
case_1_match_1_h&e_patch17	1	1	match_1	match_1	h&e	17	case_1_match_1_h&e_patch17.png	/projects/e32998/pa...	/projects/e32998/patches_varsize_poole...		4	1	F2	kimianet_pool2x2	v1	2026-11-05T...
case_1_match_1_h&e_patch18	1	1	match_1	match_1	h&e	18	case_1_match_1_h&e_patch18.png	/projects/e32998/pa...	/projects/e32998/patches_varsize_poole...		4	1	F2	kimianet_pool2x2	v1	2026-11-05T...
case_1_match_1_h&e_patch19	1	1	match_1	match_1	h&e	19	case_1_match_1_h&e_patch19.png	/projects/e32998/pa...	/projects/e32998/patches_varsize_poole...		4	1	F2	kimianet_pool2x2	v1	2026-11-05T...

- One case per patch (**83630 rows**)
- Case, slice and stain used to exist only inside name Case_1_match_1_h&e_patch0.png and every pipeline step re-parsed them. Now they are columns.
- Nothing is dropped silently, unstable patches stay in the table with a recorded reason.
- Attention maps straight back to images.

- Every image links to an embedding. Old pipeline assumed each .png had a .pt with the same name, this table will check that.

Javi

How both tables work together

Cases summary table answers the big picture questions, what's missing in the data table, and how the folds are balanced. When something looks wrong, then that is where we would go into the patches table inspect and filter more thoroughly.

EX.
Cases Table: Case 19 is missing 119 embeddings
Patches Table: df[(df.case_id == 19) & df.emb_path.isna()] list those 119 patches and their image paths, so we open then and see why.

Next Steps:

- Because 119 embeddings seems like a lot, we will make sure that everything is correct within this structure
- Add new cases when available
- Once validating everything is correct, then will start building per-case structure (one data structure per case that holds all patches metadata together).

Testing UNI2 Against KimiaNet

	KimiaNet	UNI2
Model type	DenseNet121 CNN	Vision Transformer
Size	7M parameters	681M parameters
Pretraining data	240,000+ TCGA patches	200M+ H&E and IHC images
Features per patch	4,096	1,536
Input to encoder	Patch at original size	Whole patch resized to 224×224

Kept the same for a fair test

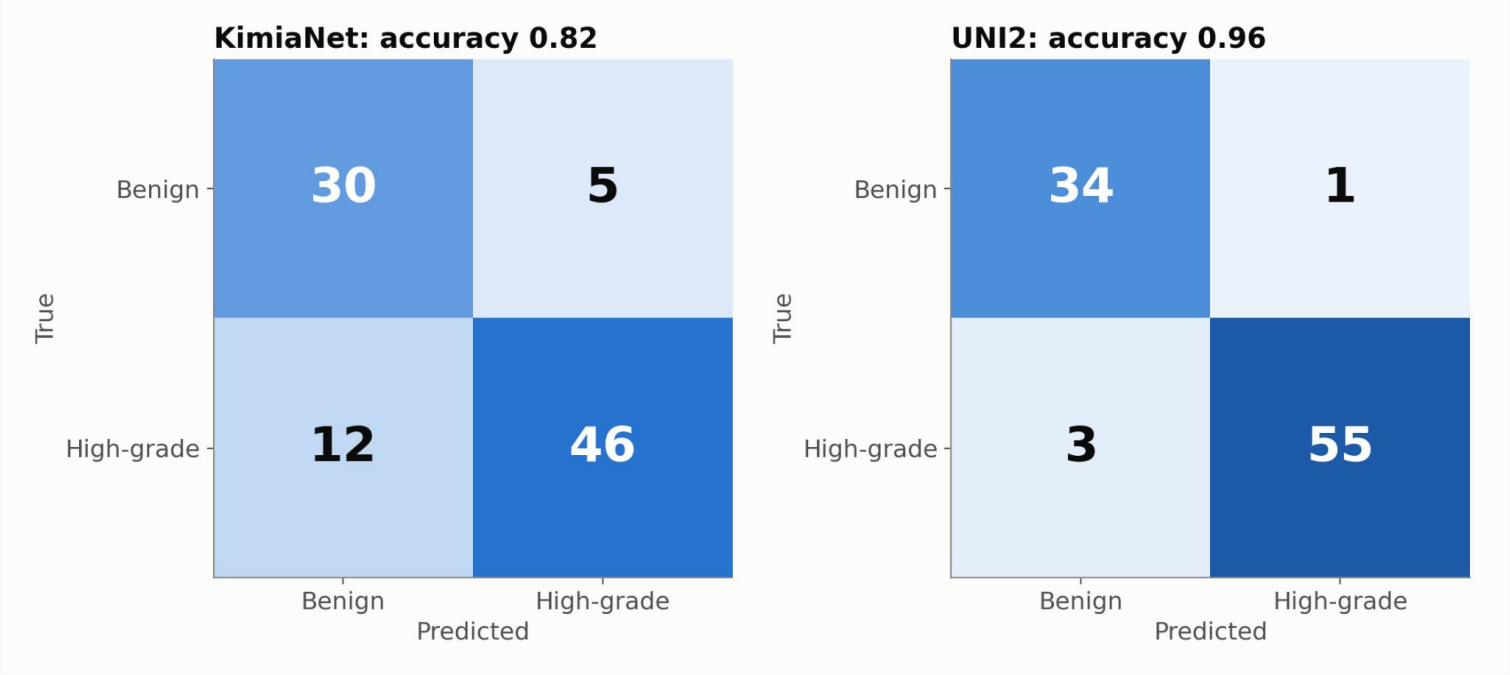
- Frozen encoders; same 5 splits; same patch → slice → stain → case attention model
- Only the projector input changed: 4,096 → 512 became 1,536 → 512
- Checkpoints chosen by validation loss; no center crop, so no part of a patch was cut off

Results

Split	Accuracy		Benign recall		High-grade recall	
	KimiaNet	UNI2	KimiaNet	UNI2	KimiaNet	UNI2
Split 1	0.737	0.842	0.714	0.857	0.750	0.833
Split 2	0.895	1.000	1.000	1.000	0.833	1.000
Split 3	0.737	1.000	0.857	1.000	0.667	1.000
Split 4	0.889	1.000	0.857	1.000	0.909	1.000
Split 5	0.833	0.944	0.857	1.000	0.818	0.909
All 93 cases	**0.817**	**0.957**	**0.857**	**0.971**	**0.793**	**0.948**

UNI2 alone was right on 14 cases; KimiaNet alone on 1 (exact McNemar p = 0.001).

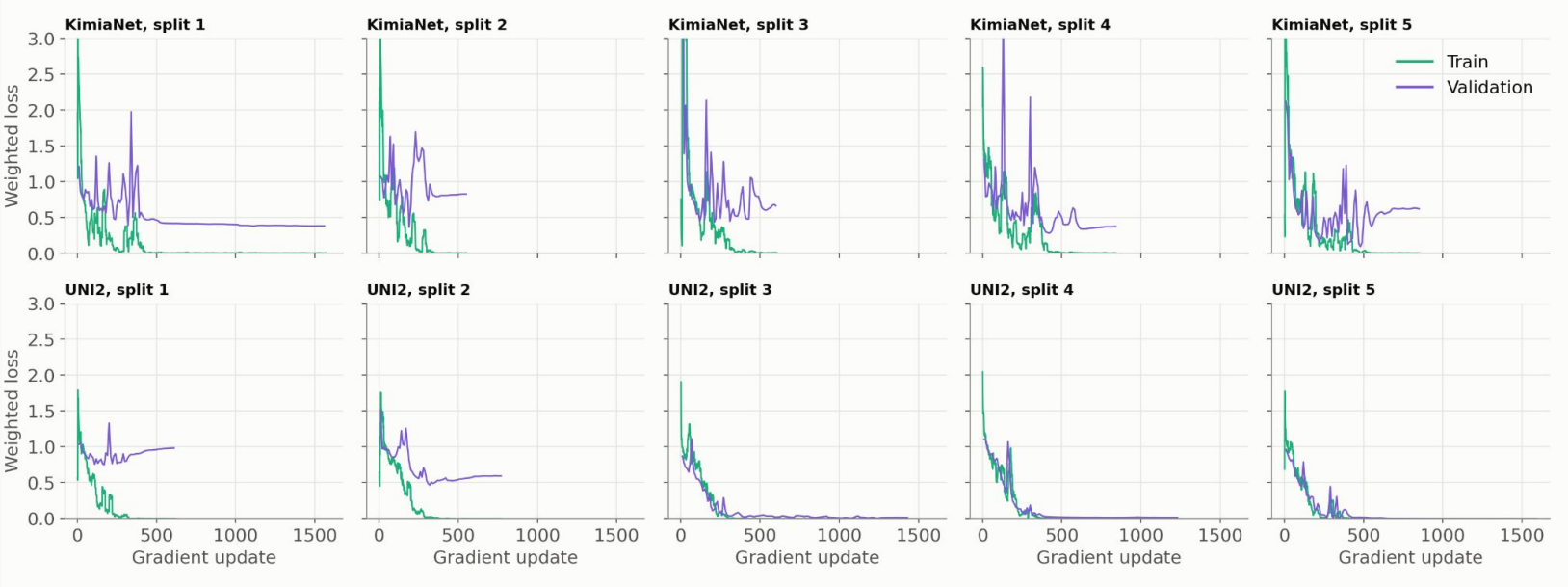
Confusion Matrices



All 93 test cases: mistakes fell from 17 with KimiaNet to 4 with UNI2.

Lucas

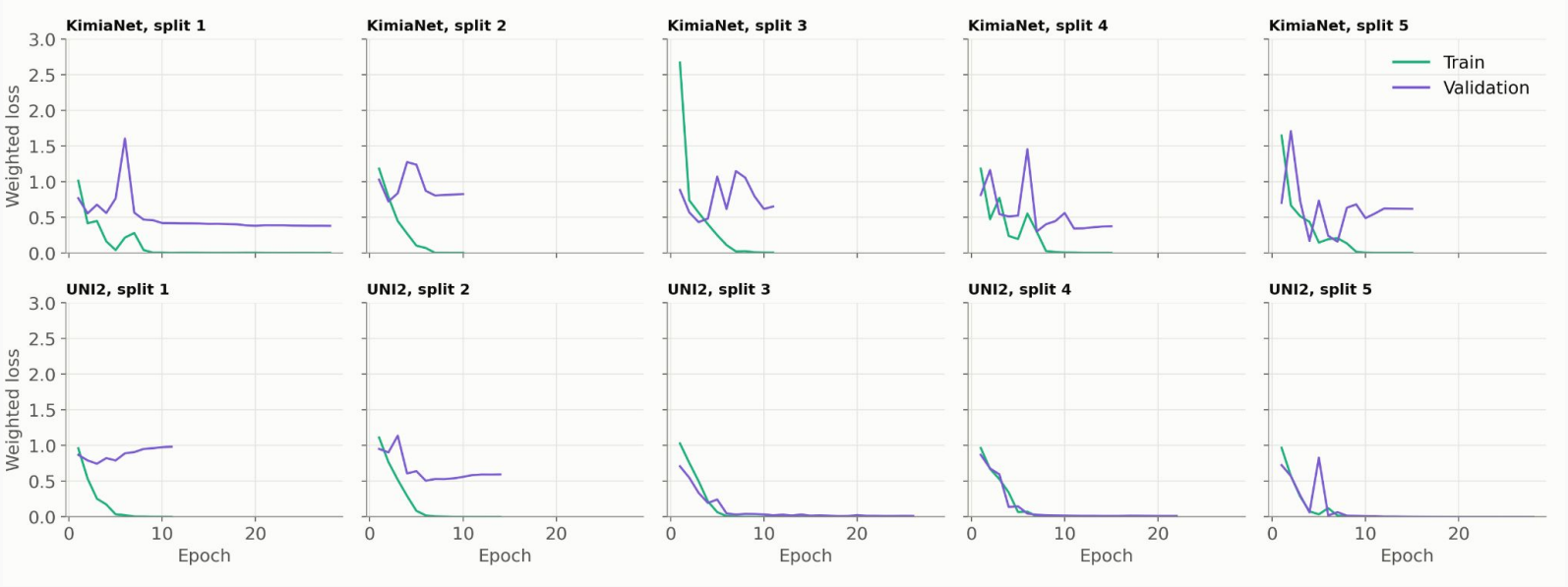
Loss per Gradient Update



KimiaNet and UNI2 splits 1–2 keep a train/validation gap; UNI2 splits 3–5 do not.

Lucas

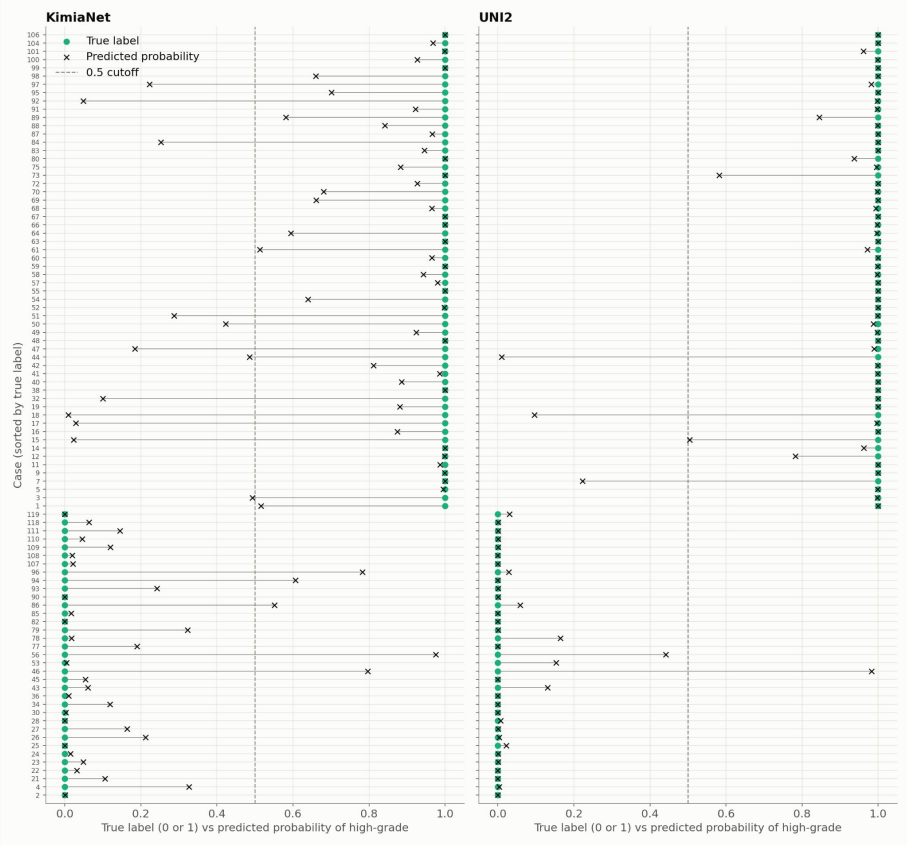
Loss per Epoch



Same pattern per epoch. Weighted loss; very early spikes above 3 are cut off.

Lucas

Prediction for Every Test Case



How to read it

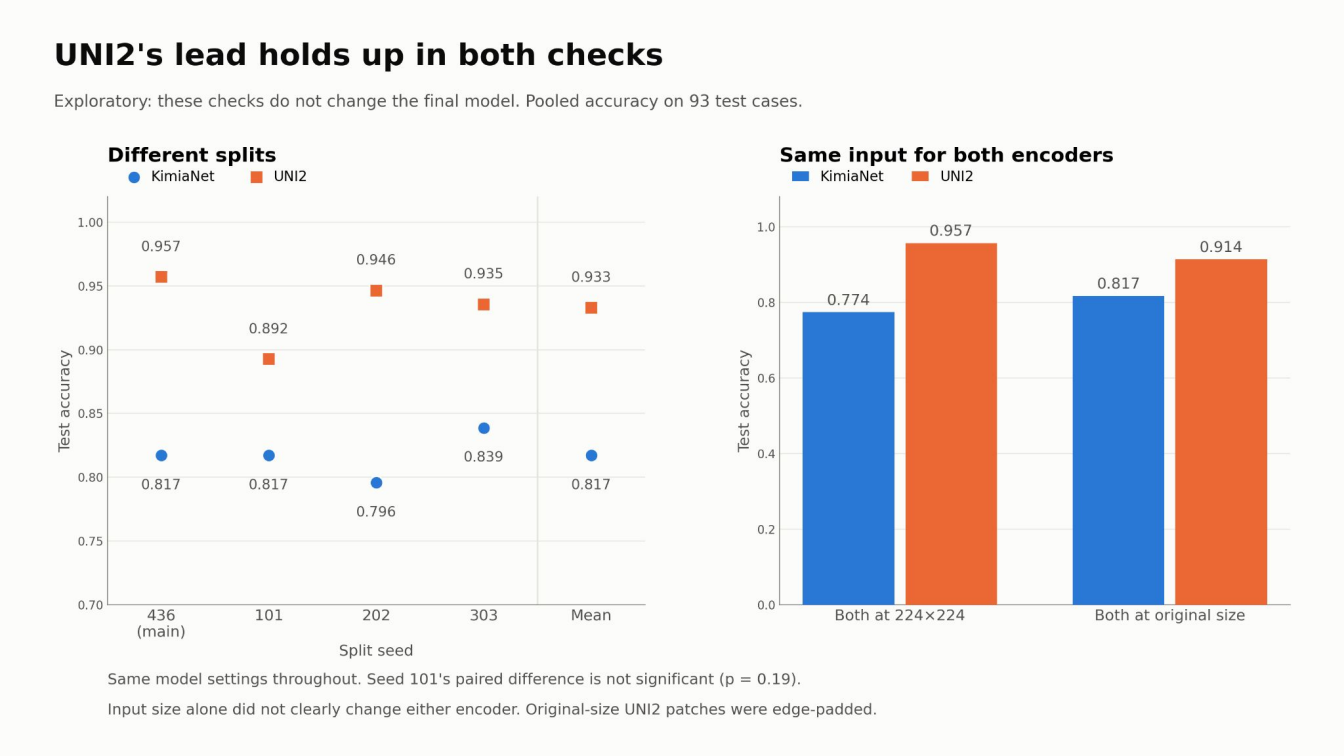
- Dot = true label; × = predicted probability of high-grade
- Longer line = bigger miss

What it shows

- KimiaNet has many long lines
- UNI2's × marks mostly sit on the dots
- UNI2 errors: cases 7, 18, 44, 46

Lucas

Exploratory Checks



Exploratory: these do not change the final model. UNI2 stayed ahead in both checks.

Lucas

Limitations and Next Steps

Limitations

- Fair comparison, but not leak-free: settings came from the earlier tuning
- 93 cases, with 18–19 test cases per split, so results are noisy
- UNI2 settings were not returned; 0.957 was the best of 4 split seeds (mean 0.933)

Suggested next steps

- Train with stains randomly hidden: UNI2 falls from 0.957 to 0.685 without H&E
- Fix the loss in optuna_training.py, then retune for UNI2 with nested CV

Class-Specific Patch Evidence

Which class does each influential patch favor?

ATTENTION

Which patches receive weight?

→

CLASS EVIDENCE

Which class does each patch favor?

A signed contribution to the Class 1 – benign score difference

$$E_{\square} = a_{\square} \times (W_1 - W_0)^T h_{\square}$$

$a_{\square}$ = stain × slice × patch attention

$h_{\square}$ = projected patch features (512D)

Positive: favors Class 1

Negative: favors benign

Andy

Correct Predictions: Opposite Directions

Top five absolute patch contributions · Melan stain, slice 0

Case 5

Class 1 · 81.33%

Patch 30	+0.1345
Patch 37	+0.1204
Patch 27	+0.1074
Patch 84	+0.0874
Patch 36	+0.0841

All five favor Class 1

Case 23

Benign · 98.68%

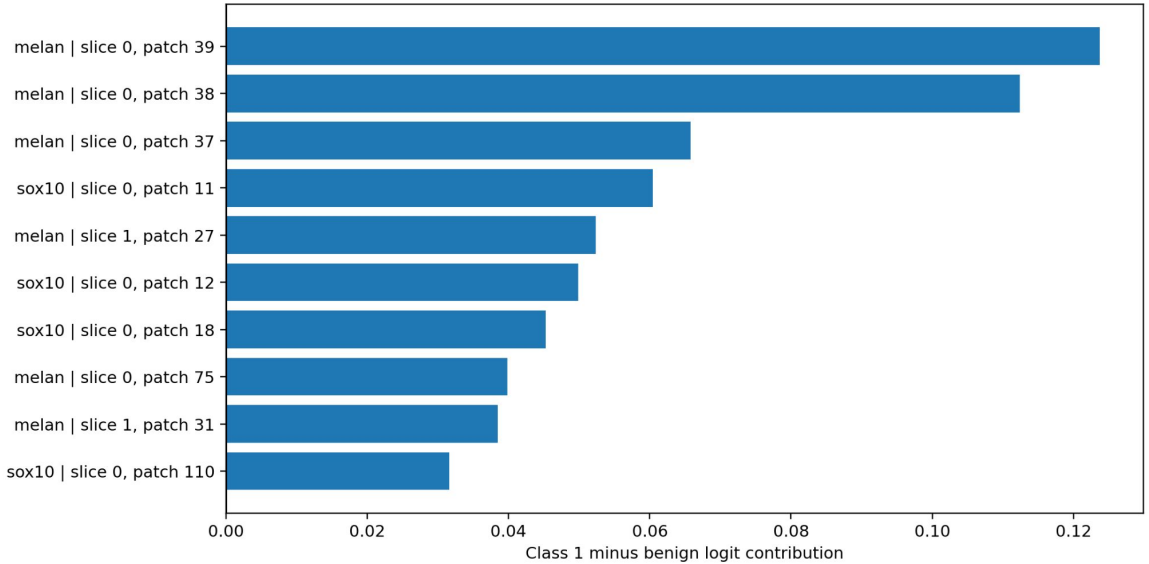
Patch 16	-0.0440
Patch 59	-0.0439
Patch 32	-0.0438
Patch 71	-0.0391
Patch 34	-0.0391

All five favor benign

Logit contributions, not probabilities · Patch numbers are indices in the tensors

Case 18: Strong Patches, Wrong Prediction

Five strongest contributions favored Class 1; model predicted benign



TRUE LABEL

Class 1

MODEL PREDICTION

Benign

66.63% probability

0 of 5

top patches favor predicted class

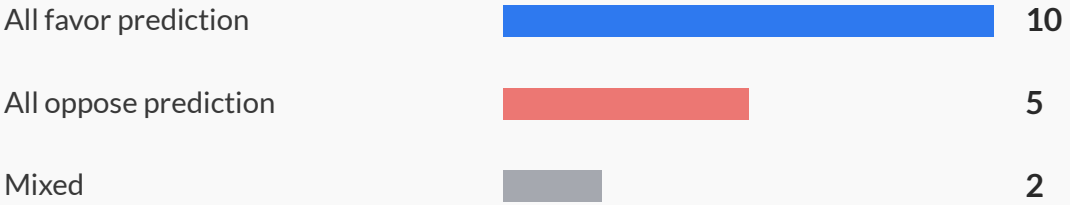
Andy

17-Case Results and Validation

Same trained MIL checkpoint · No retraining or Optuna search

17/17	14/17	13/17	4.57×10^{-7}
Predicted labels match	Probabilities match	Correct predictions	Max reconstruction error

Direction of top five patches



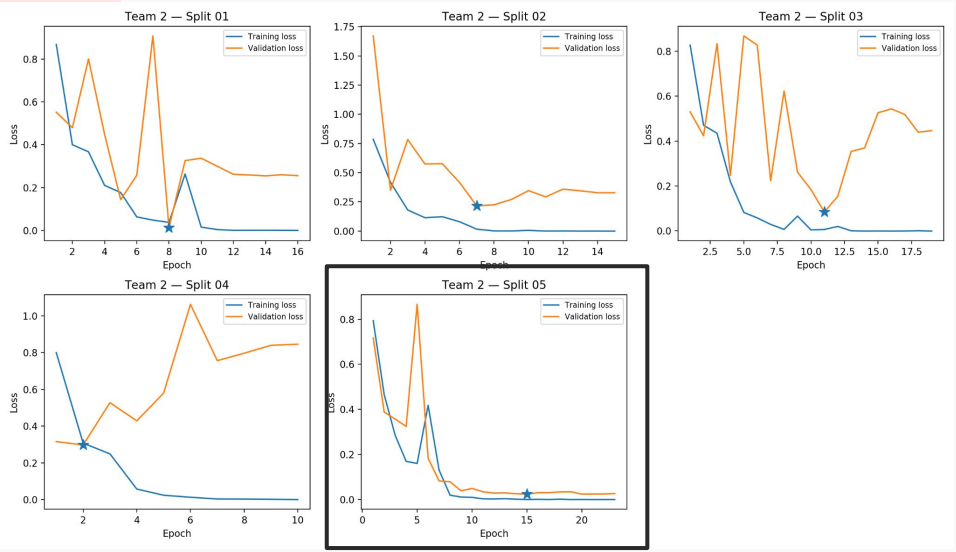
Limitations

- Not a causal patch-removal test
- 3 saved probabilities differ
- Small test set (n=17)

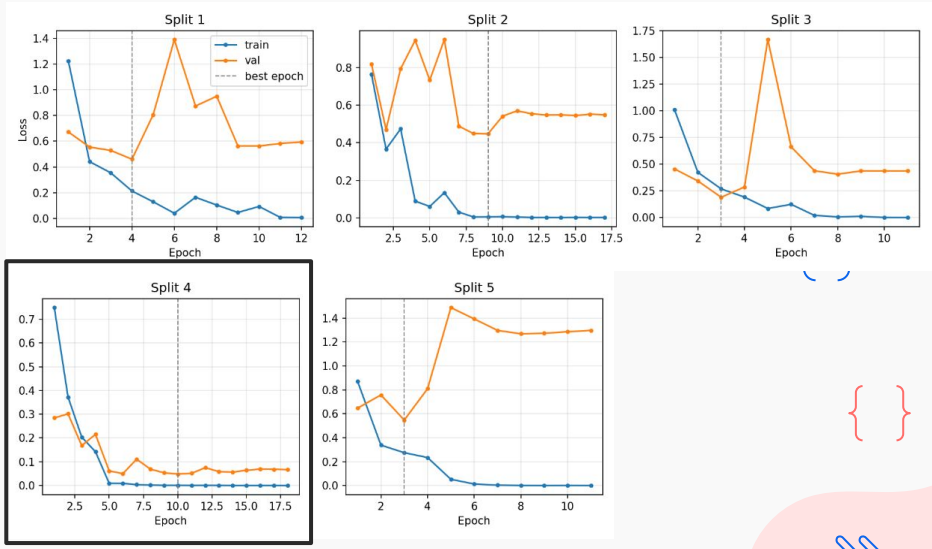
All four incorrect cases had top-five evidence opposing their prediction.

Two splits show unusually low validation loss.

Team 2 – Split 5



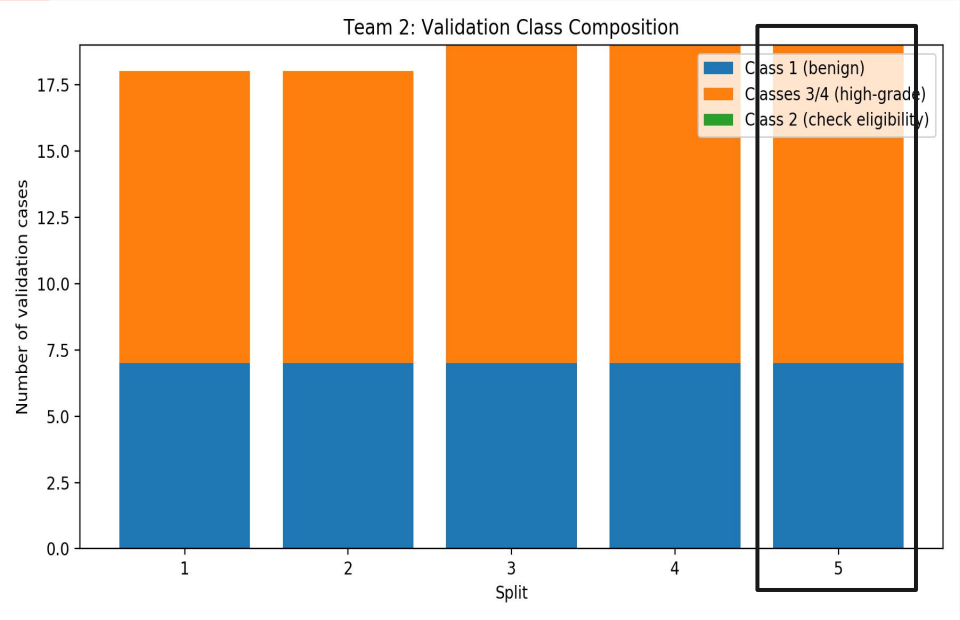
Team 5 – Split 4

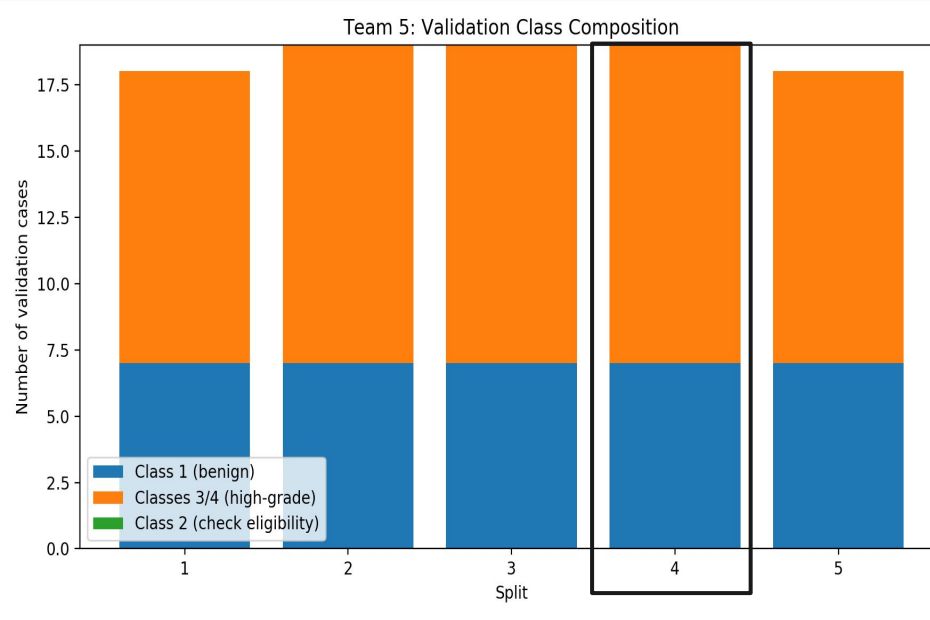


Team 2 Split 5 and Team 5 Split 4 maintain low validation loss while training loss approaches zero, unlike most other splits.

Source: Team 2, Presentation 1, analysis/loss_history.csv (visualized using analyze_losses.py); Team 5, Presentation 1, Results_230ct_Lucas/loss_curves.png

Class Balance does not explain the difference

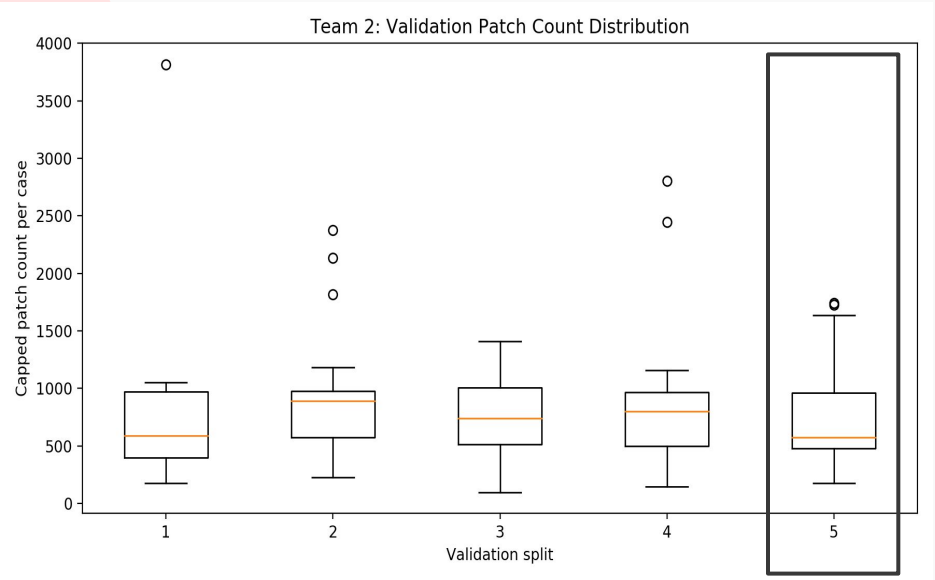


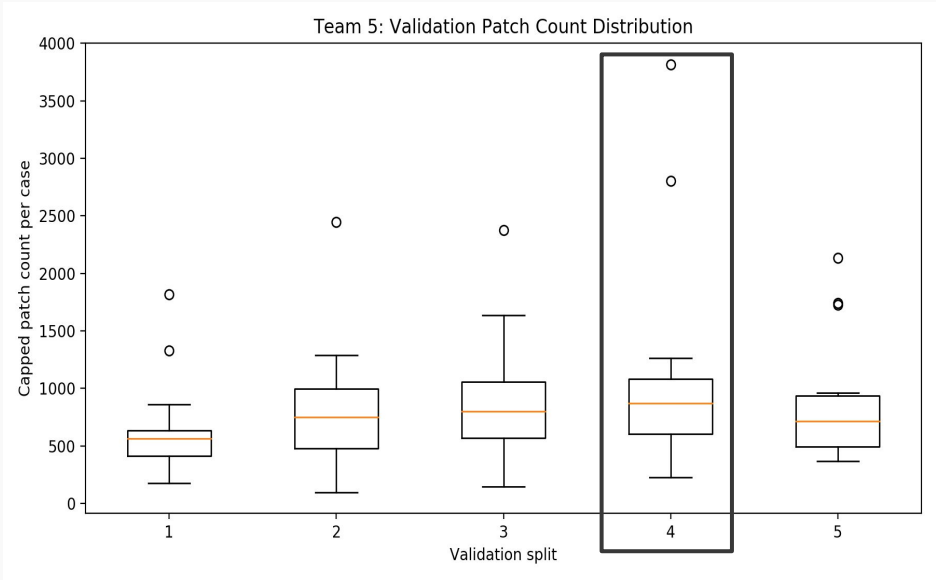


Both unusual splits contain **7 benign and 12 high-grade validation cases,** with identical training class counts (21 benign and 35 high-grade).

Source: Team 2 and Team 5, data_splits_new_01-05.npz; extra_benign_case_grade_match.csv. Figures generated using analyze_classes.py and analyze_team5_classes.py

Patch distributions tell a different story





Team 2 Split 5 has a median of 575 estimated capped patches per case.

Team 5 Split 4 has a median of 871 estimated capped patches per case - highest within Team 5.

Source: Team 2 and Team 5, data_splits_new_01-05.npz; patches_benign_split/. Figures generated using analyze_patches.py and analyze_team5_patches.py

Similar case counts, different slice composition

Measure	Team 2 - Split 5	Team 5 - Split 4
Train / validation / test	56 / 19 / 18	56 / 19 / 18
Validation benign / high-grade	7 / 12	7 / 12
Validation slice groups	56	107
Estimated capped validation patches	14,513	19,826
Median capped patches per case	575	871
Test Accuracy	94.4%	77.8%

Identical class counts but different slice and patch compositions suggest that other factors, like case difficulty or image content, may contribute to the observed validation loss behavior.

Source: Team 2 and Team 5, data_splits_new_01-05.npz; extra_benign_case_grade_match.csv; patches_benign_split/; Team 2 metrics_by_split.csv; Team 5 summary_table.csv.